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METHODS AND APPARATUS TO COLLECT MEDIA METRICS ON COMPUTING DEVICES

Abstract

Methods, apparatus, systems and articles of manufacture are disclosed that collect media metrics on computing devices. An example apparatus includes at least one memory, instructions in the apparatus, and processor circuitry to execute the instructions to determine a permission level of a meter and, in response to the determination, request a permission to access media data of an application programming interface (API), in response to determining a media session notification is available as indicated by the API, package media session data associated with the media session notification, the media session data extracted from metadata of the media session notification, obtain foreground information; and generate a package of metrics, the package of metrics including the packaged media session data and the foreground information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 18/406,259, filed Jan. 8, 2024, which is a continuation of U.S. patent application Ser. No. 17/401,124, filed Aug. 12, 2021, which issued as U.S. Pat. No. 11,895,203, and which is a continuation of U.S. patent application Ser. No. 16/368,217, filed Mar. 28, 2019, which issued as U.S. Pat. No. 11,122,134, and which claims the benefit of and priority to Indian Provisional Patent Application No. 201911005508, filed Feb. 12, 2019, each of which are hereby incorporated by reference herein in their respective entireties.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates generally to media monitoring data, and, more particularly, to methods and apparatus to collect media metrics on computing devices.

BACKGROUND

[0003] Various On Device Meters (ODM) have been used by audience measurement entities to collect data about media consumed on computing device. However, the restrictions imposed by operating systems of mobile devices (e.g., smartphones) limit the ability of ODMs to collect information. Further, there is increased interest in what users are doing within the apps (e.g., videos being watched on YouTube or Netflix, products being purchased on Amazon or songs being listened to).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a block diagram illustrating an example flow of information between the collection facility, content provider, and user device within the network.

[0005] FIG. 2 is an example block diagram of the example meter of FIG. 1 to collect and analyze media session data.

[0006] FIG. 3 is an example block diagram of the example media manager of FIG. 2 to identify and analyze session data and/or metrics.

[0007] FIG. 4 is a flowchart representative of machine readable instructions which may be executed to implement meter of FIG. 1 to obtain metrics associated with native application programming interfaces.

[0008] FIG. 5 is a flowchart representative of machine readable instructions which may be executed to implement the media manager of FIG. 2 to analyze media session notification metrics, as described above in conjunction with block 408 of FIG. 4.

[0009] FIG. 6 is a flowchart representative of machine readable instructions which may be executed to implement the audio manager of FIG. 2 to obtain audio metrics, as described above in conjunction with block 414 of FIG. 4.

[0010] FIG. 7 is a block diagram of an example processor platform structured to execute the instructions of FIGS. 4-6 to implement the meter of FIG. 1.

[0011] The figures are not to scale. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts.

[0012] Descriptors “first,” “second,” “third,” etc. are used herein when identifying multiple elements or components which may be referred to separately. Unless otherwise specified or understood based on their context of use, such descriptors are not intended to impute any meaning of priority or ordering in time but merely as labels for referring to multiple elements or components separately for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for ease of referencing multiple elements or components.

DETAILED DESCRIPTION

[0013] To obtain monitoring data (e.g., user information, user actions, duration spent performing an action) from an electronic device (e.g., a cellphone, personal laptop, personal computer, tablet, or smart watch), a meter may collect and store data (e.g., user information, user actions, duration spent performing an action). Typically, these meters are installed in the form of a downloadable application.

[0014] Prior approaches to obtaining user data via a meter installed on an electronic device include utilizing application usage data, web traffic through a local virtual private network (VPN), device meters (e.g., central processing unit (CPU) meter, battery meters, etc.) and/or radio measurement. In such prior approaches, streamed media content to the user device is unattainable. For example, content information for streamed media over Netflix™ or Hulu™ cannot be easily captured.

[0015] Examples disclosed herein utilize a combination of native Android™ application programming interfaces (APIs) to access information about media content being played, the application playing the media, and the actions taken (e.g., start, stop, pause, etc.). Example native Android™ APIs include a notification API (e.g., Android™ NotificationListenerService), a media API (e.g., Android™ MediaSession), and an audio API (e.g., Android™ AudioManager).

[0016] Examples disclosed herein include determining a permission level of the meter. In examples disclosed herein, an additional permission request is transmitted to the user. In such examples, the additional permission (e.g., notification permission) grants the meter access to the data and content generated by native Android™ APIs. Such example permission provides access to monitoring and storing media consumption on Android™ devices using Android™ native application programming interfaces (APIs).

[0017] Examples disclosed herein include utilizing the meter to obtain foreground application information and/or metrics. Such foreground application information and/or metrics include current application playing, last application playing, current time, etc. In examples disclosed herein, data obtained from native Android™ APIs is parsed with the foreground application information and/or metrics to determine similarities. For example, if the foreground application information and/or metrics provides information pertaining to the title of a video and, in parallel, data obtaining from native Android™ APIs includes a start time, a package can be created including the start time of a particular video.

[0018] The example meter implemented according to the disclosure may provide history for users

as a service. In such an example, users may observe media history across multiple applications. Additionally, the meter may log and/or otherwise record events performed by the user on the user device. Examples disclosed herein include accessing the meter on a user device to determine an action performed on the user device.

[0019] Examples disclosed herein allow the end user to obtain the meter by signing up on a third-party site (e.g., place of business or online website) and having the example meter sent to them virtually or physically, downloading the meter via a virtual app or physical memory storage device, or downloading a program via a virtual app or physical memory storage device in which the program described above contains the meter.

[0020] Examples disclosed herein include a group of panelists in which the panelists in the group are the users of the user device. These panelists are selected voluntarily or involuntarily.

Alternatively, the group of panelists may include a non-human group

[0021] FIG. 1 is a block diagram **100** illustrating an example flow of information between the collection facility **110**, content provider **108**, and user device **104** within the network **102**. The example meter **106** transmits the audience measurement data (e.g., event data) to an example collection facility **110**. The collection facility **110**, content provider **108**, and user device **104** communicate to each other via the network **102**. The meter **106**, as disclosed, is contained in the user device **104** (e.g., in the form of an application). According to the illustrated example, the meter **106** collects data (e.g., event data) about the operation of the applications on the user device **104** using the Accessibility Service.

[0022] In the illustrated example of FIG. 1, the example network **102** is a communications network. The example network **102** allows the user device **104**, the content provider **108**, and the example collection facility **110** to interface with each other to provide audience measurement data (e.g., event data) via a wired and/or wireless analog and/or digital network communication. Example wired and/or wireless analog and/or digital network communications include Zigbee, microwaves, lasers for wireless, fiber cable, communication cable, satellite communications, quantum teleportation, local area network, wide area network, personal area network, wireless local area network, the Internet, a cloud, or any other type of wired and/or wireless analog and/or digital communications network.

[0023] In the illustrated example of FIG. 1, the example user device **104** is a computing device which enables a user to connect to an example network **102** and perform actions on applications within the user device **104**. The user device **104** may include, but not limited to, a cellular phone, portable tablet, personal computer, laptop, smart watch, or gaming consoles. The example user device **104** may contain downloadable content in which the user may interact with. This data (e.g., event data) may be accessible by the meter **106** installed on the device.

[0024] In the illustrated example of FIG. 1, the example meter **106** may be in the form of a downloadable application. For example, the application may be downloaded onto the user device **104** by the device owner (e.g., the individual who is accessing content from the content provider **108** on the user device **104**) from an application store (e.g., App Store®, Google Play™, etc.). Additionally, the example meter **106** may be downloaded onto the user device **104** through the metering authority (one who is initiating the metering). For example, a user may log onto the metering authority's website and downloading from the metering authority servers. Alternatively, the meter may be downloaded via side-loading, such as USB, Wi-Fi, etc. The example meter **106** is accessible by the collection facility **110** via the network **102**. The example meter **106** measures data (e.g., event data) on the user device **104**. The data (e.g., event data) may be in the form of, but not limited to, video watched, name of video, genre, time spent on video, product viewed, or product searched for.

[0025] In the illustrated example of FIG. 1, the example installer **107** is communicatively coupled to the meter **106**. In examples disclosed herein, if the meter **106** is not installed on the user device **104**, the installer **107** can facilitate the installation of the meter **106**. Additionally, the installer **107**

determines if the meter **106** is installed on the user device **104**. If the installer **107** determines the meter **106** is not installed on the user device **104**, the installer **107** generates a request to the user device **104** to install the meter **106**. Additionally or alternatively, the installer **107** may install the meter **106** in response to a command originating by the user of the user device **104**. For example, in response to a command originating on the user device (e.g., a command from a user), the installer **107** obtains the meter **106** (e.g., from the user device **104** or the collection facility **110** via the network **102**) and executes instructions to install the meter **106** on the user device **104**. In other examples disclosed herein, the installer **107** may be implemented outside the user device **104**.

[0026] In the illustrated example of FIG. **1**, the example content provider **108** is a third-party content host which provides content (e.g., movies, songs, or news articles) to the user device **104** via the network **102**. The content provider **108** may generate, provide, or facilitate content on the user device **104**. Example content providers may include Netflix™, Hulu™, The Washington Post™, or Spotify™. The content provider **108** may provide content to the user device **104** directly through a wired connection.

[0027] In the illustrated example of FIG. **1**, the example collection facility **110** collects and manages event data from the user device **104**. The event data from the example collection facility **110** is then manipulated, organized, or sorted into measurable figures.

[0028] FIG. **2** is an example block diagram **200** of the example meter **106** of FIG. **1** to collect and analyze media session data. In the illustrated example of FIG. **2**, the meter **106** includes an example audio manager **202**, an example media manager **206**, an example detector **210**, and an example data manager **214**. The audio manager **202**, the media manager **206**, the detector **210**, and the data manager **214** communicate via the example communication bus **201**.

[0029] In the illustrated example of FIG. **2**, the communication bus **201** is a hardware conductive communication bus. In other examples disclosed herein, the communication bus **201** may be implemented via wireless communication between the audio manager **202**, the media manager **206**, the detector **210**, and the data manager **214** and/or any other suitable communication bus.

[0030] In the illustrated example of FIG. **2**, the audio manager **202** obtains data and/or metrics related to audio playing on the user device **104**. The audio manager **202** interacts with an audio API (e.g., Android™ AudioManager API) on the user device **104**. In examples disclosed herein, audio data and/or metrics refers to state changes relating to audio (e.g., changes in audio gain, changes in audio loss, changes in audio transient) that occur on the user device **104**. Additionally, audio data and/or metrics may refer to current audio states on the user device **104** such as mute, full volume, no audio playing, etc. In such examples, the audio manager **202** utilizes the permission granted when installing the meter **106**. In other examples, the audio manager **202** may send a request to the detector **210** and/or the installer **107** via the communication bus **201** to obtain a second permission to access data of any suitable Android™ API. The audio manager **202** interacts with the Android™ Audio API (e.g., the AudioManager API) to obtain a notification of an application on the user device **104** playing and/or stopping audio and/or video. In examples disclosed herein, if the audio manager **202** detects a notification from the Android™ Audio API (e.g., the AudioManager API) indicating audio and/or video on the user device **104** has started, then the audio manager **202** records the start time. Additionally, the audio manager **202** can detect a notification from the Android™ Audio API (e.g., the AudioManager API) indicating audio and/or video on the user device **104** has stopped. In such examples, the audio manager **202** records the stop time. Additionally, in such examples, the audio manager **202** can calculate the duration of the audio and package the metrics obtained in response to the audio. The audio manager **202** is coupled to an audio manager database **204**.

[0031] In the illustrated example of FIG. **2**, the audio manager database **204** stores the audio data and/or metrics obtained by the audio manager **202**. The audio manager database **204** may be physical memory located in the meter **106**. In other examples disclosed herein, the audio manager database **204** represents external storage (e.g., an external hard drive or an external cloud service)

and may be accessed wirelessly by the audio manager **202**. In examples disclosed herein, the audio manager database **204** is accessed at any time by the audio manager **202** and/or any suitable component in the meter (e.g., the media manager **206**, the detector **210**, and/or the data manager **214**).

[0032] In the illustrated example of FIG. 2, the example media manager **206** obtains data and/or metrics from the user device **104**. In examples disclosed herein, the media manager **206** obtains a notification permission from the user device **104** to access a notification API (e.g., Android™ NotificationListenerService application programming interface (API)). In other examples disclosed herein, the media manager **206** may obtain a plurality of permissions to access a plurality of APIs on the user device **104** (e.g., Android™ AudioManager API, Android™ MediaSession API, etc.). In such examples, the notification permission obtained from the user device **104** grants permission to the media manager **206** to obtain data and/or metrics (e.g., media session notifications) from a media API (e.g., the Android™ MediaSession API). In response to a user granting the requested notification permission, the media manager **206** listens for media session notifications provided by the notification API (e.g., Android™ NotificationListenerService API). In examples disclosed herein, the notification API (e.g., Android™ NotificationListenerService API) provides media metrics in conjunction with the media API (e.g., Android™ MediaSession API). For example, the media manager **206** may obtain media session data and/or metrics in response to observing a media notification through the notification API (e.g., Android™ NotificationListenerService API). The media manager **206** is explained in further detail below.

[0033] In the illustrated example of FIG. 2, the media manager database **208** stores the notification and/or media data and/or metrics obtained by the media manager **206**. The media manager database **208** may be physical memory located in the meter **106**. In other examples disclosed herein, the media manager database **208** represents external storage (e.g., an external hard drive or an external cloud service) and may be accessed wirelessly by the media manager **206**. In examples disclosed herein, the media manager database **208** is accessed at any time by the media manager **206** and/or any suitable component in the meter (e.g., the audio manager **202**, the detector **210**, and/or the data manager **214**).

[0034] In the illustrated example of FIG. 2, the detector **210** obtains foreground application metrics on the user device **104** (e.g., current active application, active application use data, etc.). The detector **210** obtains the foreground application metrics using the permission obtained when initially installing the meter **106**. For example, when a user designates permission to install the meter **106** on the user device **104**, the detector **210** may utilize such permission to obtain application foreground information. In examples disclosed herein, the detector communicates with the audio manager **202** and/or the media manager **206** to obtain respective audio and/or media data and/or metrics.

[0035] In the illustrated example of FIG. 2, the example detector database **212** stores the foreground application metrics occurring on the user device **104**. Additionally, the detector database **212** may store the notification and/or media data and/or metrics obtained by the audio manager **202** and/or the media manager **206**. The detector database **212** may be physical memory located in the meter **106**. In other examples disclosed herein, the detector database **212** represents external storage (e.g., an external hard drive or an external cloud service) and may be accessed wirelessly by the detector **210**. In examples disclosed herein, the detector database **212** is accessed at any time by the detector **210** and/or any suitable component in the meter (e.g., the audio manager **202**, the media manager **206**, and/or the data manager **214**).

[0036] In the illustrated example of FIG. 2, the example data manager **214** communicates with the audio manager **202**, the media manager **206**, and/or the detector **210** via the communication bus **201** to obtain respective audio data and/or metrics, media data and/or metrics, and/or foreground information. In examples disclosed herein, the data manager **214** parses the obtained audio data and/or metrics, media data and/or metrics, and/or obtained foreground information to create a

centralized package of metrics. For example, if the media data and/or metrics is absent of audio information (e.g., the application restricts the media API to release audio metrics), then the data manager **214** combines the similar audio data and/or metrics obtained by the audio manager **202** (e.g., through the audio API) with the media data and/or metrics and/or the foreground information to create a package of audio data and/or metrics. In examples disclosed herein, the package of data and/or metrics is stored in the data manager database **216**, accessible by the user device **104**.

[0037] In the illustrated example of FIG. **2**, the example data manager database **216** stores the package data, metrics, and/or information from the data manager **214**. The data manager database **216** may be physical memory located in the meter **106**. In other examples disclosed herein, the data manager database **216** represents external storage (e.g., an external hard drive or an external cloud service) and may be accessed wirelessly by the data manager **214**. In examples disclosed herein, the data manager database **216** is accessed at any time by the data manager **214** and/or any suitable component in the meter (e.g., the audio manager **202**, the media manager **206**, and/or the detector **210**).

[0038] FIG. **3** is an example block diagram **300** of the example media manager **206** of FIG. **2** to identify and analyze session data and/or metrics. The media manager **206** includes an example permission collector **302**, an example session identifier **304**, and an example session analyzer **306**. In the example illustrated in FIG. **3**, the permission collector **302**, the session identifier **304**, and/or the session analyzer communicate via an example media communication bus **301**.

[0039] In the illustrated example of FIG. **3**, the media communication bus **301** is a hardware conductive communication bus. In other examples disclosed herein, the media communication bus **301** is illustrative of wireless communication between the permission collector **302**, the session identifier **304**, and/or the session analyzer **306**.

[0040] In the illustrated example of FIG. **3**, the example permission collector **302** obtains and/or otherwise manages available permissions for the media manager **206**. In examples disclosed herein, the permission collector **302** determines if the permission level of the meter **106** (e.g., determines if the desired media permission is granted). If the permission collector **302** determines the desired media permission is not granted, the permission collector **302** can facilitate the request for the desired media permission to access media data. In some examples disclosed herein, the permission collector **302** may transmit a request indication to the user of the user device **104** (e.g., transmit the request indication via the media communication bus **301** and further via the communication bus **201** of FIG. **2**) to grant the desired media permission. In such examples, the user of the user device **104** is prompted to provide permission to the permission collector **302** to access data and/or metrics from the notification API (e.g., Android™ NotificationListenerService). In examples disclosed herein, the permission obtained to access data and/or metrics from the notification API may be the same permission to access data and/or metrics from the media API (e.g., Android™ MediaService). Additionally or alternatively, if an additional permission to access data and/or metrics in the user device, the permission collector **302** can facilitate the request and reception of such additional permission (e.g., additional permission to access alternative APIs, the media API, etc.).

[0041] In the illustrated example of FIG. **3**, the example session identifier **304** listens to the notification API (e.g., Android™ NotificationListener) to determine notifications that occur. Example notifications observable by the session identifier **304** that originate from the notification API in the user device **104** include any calls from the user device **104** when a new notification is posted, removed, or their ranking is changed. For example, if a video is being played on the user device **104**, the notification API generates a notification event stating the change, such notification event is observable by the media manager **206** of FIG. **2**, and, more particularly, the session identifier **304**. In examples disclosed herein, the session identifier **304** listens to the media API (e.g., Android™ MediaService) in parallel with the notification API (e.g., Android™ NotificationListener) to obtain media data and/or metrics corresponding to the observance of a notification. For example, if a video is being played on the user device **104**, the session identifier

304 listens and/or otherwise observes the corresponding notification and similarly listens and/or otherwise observes for the occurrence of respective media data and/or metrics. Example media data and/or metrics include an associated MediaSession, an associated ActiveSession, etc.

[0042] In the example illustrated in FIG. 3, the session analyzer **306** analyzes and/or otherwise interprets the observed notifications from the session identifier **304**. In response to a media data and/or metrics associated with the observed notification being available, the example session analyzer **306** determines if the media data and/or metrics includes an active media session (e.g., ActiveSession). If an active media session (e.g., ActiveSession) is available, the session analyzer **306** obtains a list media track functions (e.g., mediaController) and/or associated active media session data. The session analyzer **306** extracts the relevant metadata and/or list of media track functions (e.g., mediaController functions) associated with the media data and/or metrics. For example, after the session identifier **304** identifies a notification from the notification API (e.g., Android™ NotificationListenerService) and whether the notification includes relevant media data and/or metrics (e.g., MediaSession), the session analyzer **306** determines if the media data and/or metrics (e.g., MediaSession) is active media data and/or metrics (e.g., ActiveSession). In such examples, the session analyzer **306** extracts a relevant list of media track functions (e.g., mediaController functions) and corresponding metadata (e.g., title, application name, etc.). As such, the session analyzer **306** creates a package of media session data using the extracted data and/or metrics, the packaged media session data for use by the meter **106**.

[0043] Additionally or alternatively, if an active media session (e.g., ActiveSession) is not available, the session analyzer **306** obtains the applicable media session playback data available from the media data and/or metrics (e.g., MediaSession). The session analyzer **306** extracts the applicable media session playback data associated with the media data and/or metrics (e.g., MediaSession). For example, after the session identifier **304** identifies a notification from the notification API (e.g., Android™ NotificationListenerService) and whether the notification includes relevant media data and/or metrics (e.g., MediaSession), the session analyzer **306** determines if the media data and/or metrics (e.g., MediaSession) is active media data and/or metrics (e.g., ActiveSession). In such examples, the session analyzer **306** extracts applicable media session playback data. In such examples, since active media data and/or metrics are not available, the session analyzer **306** generates an indication to request audio data and/or metrics. As such, the session analyzer **306** creates a package of available media session data using the extracted data and/or metrics, the available packaged media session data for use by the meter **106**.

[0044] While an example manner of implementing the meter **106** of FIG. 1 is illustrated in FIGS. 2 and 3, one or more of the elements, processes and/or devices illustrated in FIGS. 2 and 3 may be combined, divided, re-arranged, omitted, eliminated and/or implemented in any other way. Further, the example audio manager **202**, the example media manager **206**, the example detector **210**, the example data manager **214**, the example permission collector **302**, the example session identifier **304**, and the example session analyzer **306** and/or, more generally, the example meter **106** of FIG. 1 may be implemented by hardware, software, firmware and/or any combination of hardware, software and/or firmware. Thus, for example, any of the example audio manager **202**, the example media manager **206**, the example detector **210**, the example data manager **214**, the example permission collector **302**, the example session identifier **304**, and the example session analyzer **306** and/or, more generally, the example meter **106** of FIG. 1 could be implemented by one or more analog or digital circuit(s), logic circuits, programmable processor(s), programmable controller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)) and/or field programmable logic device(s) (FPLD(s)). When reading any of the apparatus or system claims of this patent to cover a purely software and/or firmware implementation, at least one of the example audio manager **202**, the example media manager **206**, the example detector **210**, the example data manager **214**, the example permission collector **302**, the example session identifier **304**, and the

example session analyzer **306** is/are hereby expressly defined to include a non-transitory computer readable storage device or storage disk such as a memory, a digital versatile disk (DVD), a compact disk (CD), a Blu-ray disk, etc. including the software and/or firmware. Further still, the example meter **106** of FIG. **1** may include one or more elements, processes and/or devices in addition to, or instead of, those illustrated in FIGS. **2** and **3**, and/or may include more than one of any or all of the illustrated elements, processes and devices. As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

[0045] A flowchart representative of example hardware logic, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing the meter **106** of FIG. **1** is shown in FIGS. **2** and **3**. The machine readable instructions may be one or more executable programs or portion(s) of an executable program for execution by a computer processor such as the processor **712** shown in the example processor platform **700** discussed below in connection with FIG. **7**. The program may be embodied in software stored on a non-transitory computer readable storage medium such as a CD-ROM, a floppy disk, a hard drive, a DVD, a Blu-ray disk, or a memory associated with the processor **712**, but the entire program and/or parts thereof could alternatively be executed by a device other than the processor **712** and/or embodied in firmware or dedicated hardware. Further, although the example program is described with reference to the flowchart illustrated in FIGS. **4-6**, many other methods of implementing the example meter **106** may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware.

[0046] The machine readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a packaged format, etc. Machine readable instructions as described herein may be stored as data (e.g., portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine readable instructions may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers). The machine readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, etc. in order to make them directly readable and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and stored on separate computing devices, wherein the parts when decrypted, decompressed, and combined form a set of executable instructions that implement a program such as that described herein. In another example, the machine readable instructions may be stored in a state in which they may be read by a computer, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc. in order to execute the instructions on a particular computing device or other device. In another example, the machine readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, the disclosed machine readable instructions and/or corresponding program(s) are intended to encompass such machine readable instructions and/or program(s) regardless of the particular format or state of the machine readable instructions and/or program(s)

when stored or otherwise at rest or in transit.

[0047] As mentioned above, the example processes of FIGS. **4-6** may be implemented using executable instructions (e.g., computer and/or machine readable instructions) stored on a non-transitory computer and/or machine readable medium such as a hard disk drive, a flash memory, a read-only memory, a compact disk, a digital versatile disk, a cache, a random-access memory and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the term non-transitory computer readable medium is expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media.

[0048] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc. may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

[0049] FIG. **4** is a flowchart **400** representative of machine readable instructions which may be executed to implement meter **106** of FIG. **1** to obtain metrics associated with native application programming interfaces. Illustrated in FIG. **4**, the permission collector **302** of the media manager **206** determines if the desired permission is granted (block **402**). If the desired permission is not granted, the permission collector **302** of the media manager **206** requests the desired permission (block **404**). In examples disclosed herein, the desired permission provides access to a notification API (e.g., Android™ NotificationListenerService).

[0050] If the desired permission is granted, the example session identifier **304** of the media manager **206** determines if a media session notification is available (block **406**). If a media session notification is not available, then control is directed to the audio manager **202** of the meter **106** to obtain audio metrics (block **414**). Example control executed by the audio manager **202** of the meter **106** to obtain audio metrics is explained in further detail below. Alternatively, if a media session notification is available, the session analyzer **306** of the media manager **206** analyzes the media session notification metrics (block **408**). Example control executed by the session analyzer **306** to analyze the media session notification metrics is explained in further detail below.

[0051] In response, the session analyzer **306** obtains the analyzed package (block **410**). In examples disclosed herein, the analyzed package obtained may include general media session playback data and/or metadata related to the media session notification. Additionally, in response to the analyzed

package obtained, the session analyzer **306** determines whether audio metrics are requested (block **412**). If audio metrics are requested (e.g., the obtained analyzed package does not obtain metadata), control returns to the audio manager **202** to execute the function of block **414**.

[0052] In response to either obtaining audio metrics (block **414**) or determining audio metrics are not requested (e.g., control response negative to executing the function of block **412**), then the detector **210** of the meter **106** obtains foreground application metrics (block **416**). The data manager **214** of the meter **106** parses the obtained metrics (e.g., the media session notification metrics and/or the audio metrics) with obtained foreground application metrics (e.g., the obtained foreground application metrics from block **416**) (block **418**). In response to parsing the abovementioned data, the data manager **214** packages the corresponding metrics (block **420**). In examples disclosed herein, the packaged metrics from block **420** include relevant data relating to the media session notification identified in block **406**.

[0053] In examples disclosed herein, the meter **106** and, thus, the audio manager **202**, the media manager **206**, the detector **210**, and/or the data manager **214** determine whether to continue operating (block **422**). Examples in which the meter **106** determines to cease operation include loss of power, loss of general permission (e.g., the permission to operate the meter **106** is lost), failure to operate, etc. In examples in which the meter **106** determines to continue operating, control returns to block **406** for the session identifier **304** of the media manger **206** to determine if a media session notification is available (block **406**).

[0054] FIG. **5** is a flowchart representative of machine readable instructions which may be executed to implement the media manager **206** of FIG. **2** to analyze media session notification metrics, as described above in conjunction with block **408** of FIG. **4**. The media manger **206** determines if active media session metrics are included in the media session notification (block **502**). If active media session metrics are included, the media manger **206** extracts the metadata associated with the media session notification (block **504**). In addition, the media manager **206** packages the metadata (block **506**).

[0055] If active media session metrics are not included, the media manger **206** extracts media session playback data (block **508**). In examples disclosed herein, the media session playback data includes relevant media session information such as play time and/or stop time. The media manager **206** packages the media session playback data (block **510**).

[0056] FIG. **6** is a flowchart representative of machine readable instructions which may be executed to implement the audio manager **202** of FIG. **2** to obtain audio metrics, as described above in conjunction with block **414** of FIG. **4**. The audio manager **202** determines if an audio start is detected (block **602**). If an audio start is not detected, then the audio manger **202** determines if audio is currently being played (block **604**). In response, if the audio manager **202** determines audio is not currently being played, the audio manager **202** identifies that no audio data is available (block **622**).

[0057] Alternatively, in response to the control executed in block **604**, if it is determined that audio data is currently playing, the audio manager **202** determines if current audio is requested (block **606**). In executing the control of block **606**, if the audio manager **202** determines the current audio data is requested, the audio manager **202** packages the known audio data (e.g., the known duration of audio playing) (block **624**). If the audio manager **202** determines the current audio data is not requested, the audio manager **202** determines if an audio stop is detected (block **608**).

[0058] Returning to the control executed by the audio manger **202** in block **602**, if an audio start is detected, then the audio manager **202** records the audio start time (block **616**). In response, the audio manager **202** determines if the current audio data is requested (block **618**). In examples disclosed herein, the control executed by the audio manager **202** in block **618** may be the same control executed by the audio manager in block **606**. In executing the control of block **618**, if the audio manager **202** determines the current audio data is requested, the audio manager **202** packages the known audio data (e.g., the known audio start time) (block **620**). In examples disclosed herein,

the control executed by the audio manager **202** in block **620** may be the same control executed by the audio manager in block **624**. If the audio manager **202** determines the current audio data is not requested, the audio manager **202** determines if an audio stop is detected (block **608**).

[0059] If an audio stop is not detected, the audio manager **202** checks to determine if the current audio data is requested (block **618**). In other examples disclosed herein, in response to an audio stop not detected, the audio manager **202** may return to the control executed in block **606**.

[0060] In examples disclosed herein in which an audio stop is detected, the audio manager **202** records the audio stop time (block **610**). The audio manager **202** determines the known audio duration (e.g., the time elapsed between the audio start time and the audio stop time, or the time elapsed between detecting audio currently playing and the audio stop time) (block **612**). In response, the audio manager **202** packages the audio duration and corresponding data (block **614**).

[0061] FIG. 7 is a block diagram of an example processor platform **700** structured to execute the instructions of FIGS. 4-6 to implement the meter **106** of FIG. 1. The processor platform **700** can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), a mobile device (e.g., a cell phone, a smart phone, a tablet such as an iPad™), a personal digital assistant (PDA), an Internet appliance, a DVD player, a CD player, a digital video recorder, a Blu-ray player, a gaming console, a personal video recorder, a set top box, a headset or other wearable device, or any other type of computing device.

[0062] The processor platform **700** of the illustrated example includes a processor **712**. The processor **712** of the illustrated example is hardware. For example, the processor **712** can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUs, DSPs, or controllers from any desired family or manufacturer. The hardware processor may be a semiconductor based (e.g., silicon based) device. In this example, the processor implements the example audio manager **202**, the example media manager **206**, the example detector **210**, the example data manager **214**, the example permission collector **302**, the example session identifier **304**, and the example session analyzer **306**.

[0063] The processor **712** of the illustrated example includes a local memory **713** (e.g., a cache). The processor **712** of the illustrated example is in communication with a main memory including a volatile memory **714** and a non-volatile memory **716** via a bus **718**. The volatile memory **714** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of random access memory device. The non-volatile memory **716** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **714**, **716** is controlled by a memory controller.

[0064] The processor platform **700** of the illustrated example also includes an interface circuit **720**. The interface circuit **720** may be implemented by any type of interface standard, such as an Ethernet interface, a universal serial bus (USB), a Bluetooth® interface, a near field communication (NFC) interface, and/or a PCI express interface.

[0065] In the illustrated example, one or more input devices **722** are connected to the interface circuit **720**. The input device(s) **722** permit(s) a user to enter data and/or commands into the processor **712**. The input device(s) can be implemented by, for example, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, isopoint and/or a voice recognition system.

[0066] One or more output devices **724** are also connected to the interface circuit **720** of the illustrated example. The output devices **724** can be implemented, for example, by display devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube display (CRT), an in-place switching (IPS) display, a touchscreen, etc.), a tactile output device, a printer and/or speaker. The interface circuit **720** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or a graphics driver processor.

[0067] The interface circuit **720** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) via a network **726**. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, etc.

[0068] The processor platform **700** of the illustrated example also includes one or more mass storage devices **728** for storing software and/or data. Examples of such mass storage devices **728** include floppy disk drives, hard drive disks, compact disk drives, Blu-ray disk drives, redundant array of independent disks (RAID) systems, and digital versatile disk (DVD) drives.

[0069] The machine executable instructions **732** of FIGS. **4-6** may be stored in the mass storage device **728**, in the volatile memory **714**, in the non-volatile memory **716**, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

[0070] From the foregoing, it will be appreciated that example methods, apparatus and articles of manufacture have been disclosed that collect user media measurement data through native application programming interfaces. The disclosed methods, apparatus and articles of manufacture improve the efficiency of using a computing device by obtaining media metrics for streamed media on a user device. The disclosed methods, apparatus and article of manufacture utilize less processing power by parsing media metrics from a variety of native application programming interfaces to obtain reliable media metric data. As such, reliable media metric data can be obtained without having to perform redundant processing tasks. The disclosed methods, apparatus and article of manufacture utilize native APIs in a manner that allows for efficient collection of media metrics by combining the native API media metric data with foreground application data. The disclosed methods, apparatus and articles of manufacture are accordingly directed to one or more improvement(s) in the functioning of a computer.

[0071] Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. An audience measurement meter associated with a media playback device, the audience measurement meter comprising: a processor; and memory having stored thereon computer readable instructions that, when executed by the processor, cause the audience measurement meter to perform operations comprising: based on receiving permission to access notifications from both (i) a first application programming interface (API) and (ii) a second API, obtaining, via the second API, a current audio playback state of the media playback device; obtaining, via the second API, an audio playback start time of audio content being played back by the media playback device; obtaining a current application being executed in a foreground of the media playback device; and reporting indications of the audio playback start time and the current application.
2. The audience measurement meter of claim 1, wherein the operations further include determining a duration of audio playback of the audio content based on the audio playback start time.
3. The audience measurement meter of claim 1, wherein the operations further include: obtaining, via the second API, an audio playback stop time of the audio content being played back by the media playback device; and determining a duration of audio playback of the audio content based on the audio playback start time and the audio playback stop time.
4. The audience measurement meter of claim 1, wherein the operations further include receiving, via the first API, a media session notification indicative of an initiation of a media playback session

on the media playback device; and wherein the obtaining the current audio playback state of the media playback device is based on receiving the media session notification indicative of the initiation of the media playback session.

5. The audience measurement meter of claim 1, wherein receiving permission to access notifications from both the first API and the second API includes requesting, from at least one of the first API or the second API, permission to access data of the at least one of the first API or the second API.

6. The audience measurement meter of claim 5, wherein requesting permission to access data of the at least one of the first API or the second API is responsive to determining that a permission level of the audience measurement meter does not permit access to notifications of the at least one of the first API or the second API.

7. The audience measurement meter of claim 1, wherein the media playback device and the audience measurement meter are implemented in a single hardware device.

8. A non-transitory computer readable storage medium having stored thereon instructions that, when executed by a processor of a media playback device, cause performance of a set of operations comprising: based on receiving permission to access notifications from both (i) a first application programming interface (API) and (ii) a second API, obtaining, via the second API, a current audio playback state of the media playback device; obtaining, via the second API, an audio playback start time of audio content being played back by the media playback device; obtaining a current application being executed in a foreground of the media playback device; and reporting indications of the audio playback start time and the current application.

9. The non-transitory computer readable storage medium of claim 8, wherein the operations further include determining a duration of audio playback of the audio content based on the audio playback start time.

10. The non-transitory computer readable storage medium of claim 8, wherein the operations further include: obtaining, via the second API, an audio playback stop time of the audio content being played back by the media playback device; and determining a duration of audio playback of the audio content based on the audio playback start time and the audio playback stop time.

11. The non-transitory computer readable storage medium of claim 8, wherein the operations further include: obtaining foreground information, wherein the current application being executed is indicated by the foreground information; identifying a title of audio content being played back by the media playback device based on the foreground information; and reporting the title in association with the audio playback start time and the current application.

12. The non-transitory computer readable storage medium of claim 8, wherein receiving permission to access notifications from both the first API and the second API includes requesting, from at least one of the first API or the second API, permission to access data of the at least one of the first API or the second API.

13. The non-transitory computer readable storage medium of claim 12, wherein requesting permission to access data of the at least one of the first API or the second API is responsive to determining that a permission level of the media playback device does not permit access to notifications of the at least one of the first API or the second API.

14. A method comprising: based on receiving permission to access notifications from both (i) a first application programming interface (API) and (ii) a second API, obtaining, via the second API, a current audio playback state of a media playback device; obtaining, via the second API, an audio playback start time of audio content being played back by the media playback device; obtaining a current application being executed in a foreground of the media playback device; and reporting indications of the audio playback start time and the current application.

15. The method of claim 14, further comprising determining a duration of audio playback of the audio content based on the audio playback start time.

16. The method of claim 14, further comprising: obtaining, via the second API, an audio playback

stop time of the audio content being played back by the media playback device; and determining a duration of audio playback of the audio content based on the audio playback start time and the audio playback stop time.

17. The method of claim 14, further comprising receiving, via the first API, a media session notification indicative of an initiation of a media playback session on the media playback device; and wherein the obtaining the current audio playback state of the media playback device is based on receiving the media session notification indicative of the initiation of the media playback session.

18. The method of claim 14, further comprising: obtaining foreground information, wherein the current application being executed is indicated by the foreground information; identifying a title of audio content being played back by the media playback device based on the foreground information; and reporting the title in association with the audio playback start time and the current application.

19. The method of claim 14, wherein receiving permission to access notifications from both the first API and the second API includes requesting, from at least one of the first API or the second API, permission to access data of the at least one of the first API or the second API.

20. The method of claim 19, wherein requesting permission to access data of the at least one of the first API or the second API is responsive to determining that a permission level of the media playback device does not permit access to notifications of the at least one of the first API or the second API.
